

19	B	(A) is wrong: $E = \frac{Q}{4\pi\epsilon_0 r^2},$ as r increases, E will decrease. (B) is correct:

Qn	Ans	Solution
		$U = \frac{Q_1 Q_2}{4\pi\epsilon_0 r}, U \text{ is negative. As } r \text{ increases, } U \text{ becomes less negative.}$ U increases (C) is wrong. Electric force is always acting on the moving positive charge, perpendicular to the equipotential line ST. (D) is wrong. Force by external agent is in the opposite direction to the displacement, so work done will be negative
20	A	Let current in the circuit be I .